Answer Key - Mathematics - K-12 Level Assessment 3 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** C
   Options: A) 16; B) 20; C) 24; D) 28
   Note: Correct option: C - 24
2. **Answer:** A
   Options: A) 4.5; B) 2.25; C) 6.6; D) 3.3
   Note: Correct option: A - 4.5
3. **Answer:** C
   Options: A) 100; B) 999; C) 625; D) 500
   Note: Correct option: C - 625
4. **Answer:** B
   Options: A) central tendency.; B) variability.; C) symmetry.; D) skewness.
   Note: Correct option: B - variability.
5. **Answer:** D
   Options: A) 5; B) 11; C) 17; D) 199
   Note: Correct option: D - 199

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '1 and 1 over 2 ft'.
7. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.
8. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '7'.
9. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.
10. **Answer:** False
    Options: A) True; B) False
    Note: LLM-generated false statement. Correct answer: '5.7'.

Long Form Response

11. **Answer:** Note that
$$a_k = \frac{1}{k^2+k} = \frac{1}{k} - \frac{1}{k+1}$$
for each k. Thus, the sum telescopes:
$$a_m + a_{m+1} + \ldots + a_{n-1} = \left(\frac{1}{m} - \frac{1}{m+1}\right) + \left(\frac{1}{m+1} - \frac{1}{m+2}\right) + \ldots + \left(\frac{1}{n-1} - \frac{1}{n}\right) = \frac{1}{m} - \frac{1}{n}.$$
Therefore, we have the equation $1/m - 1/n = 1/29$. Multiplying by $29mn$ on both sides, we have $29n - 29m = mn$, or $mn + 29m - 29n = 0$. Subtracting 29^2 from both sides, we get
$$(m-29)(n+29) = -29^2.$$
Since 29 is prime and $0 < m < n$, the only possibility is $m - 29 = -1$ and $n + 29 = 841$, which gives $m = 28$ and $n = 812$. Thus, $m + n = 28 + 812 = 840$.
Note: Students should show all steps in their mathematical solution.

12. **Answer:** $2 \leq \mathrm{y} \leq 8$. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key